Michael Yang

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Education

Carleton University Ottawa, ON

Bachelor of Computer Science, Honours in AI and ML, Co-op Option *Sept. 2023 – Apr 2028*

- **Relevant Coursework**: Abstract Data Types and Algorithms, Discrete Structures II, Introduction to Systems Programming, Introduction to Software Engineering

Technical Skills

Languages: Java, Python, C, C++, TypeScript, JavaScript, HTML/CSS, Lua, C#, Go
Frameworks/Libraries: React.js, Node.js, Pytest, Flask, TailwindCSS, LangChain, Express.js, streamlit, Next.js, agno
Developer Tools: Git, Docker, MongoDB, Github, Jupyter, Confluence, Jira, TestRail, Prometheus, Grafana, Github Actions, Kubernetes

Experience

Software Developer Co-op - AI Summer 2026

JSI Telecom *Ottawa, ON*

- Implemented an automated **LLM** benchmarking **CI** pipeline using **k6** and **GitHub Actions**, simulating **50+** concurrent virtual users generating **80000+** total requests to track model latency and throughput on a **Grafana** dashboard.
- Scaled **container pods** to accommodate increasing virtual user traffic against a **8-million-entry** dataset, reducing bottlenecks and reducing crash rates by **99%**.

Software Developer Co-op - AI Sept 2025 – Dec 2025

JSI Telecom *Ottawa, ON*

- Developed task-specific LLM agents utilizing **Python**, **LanceDB**, and **Agno**, designing an FAQ agent and prompt alignment verifier, while utilizing **DeepEval** to maintain a **95%+** output consistency rate.
- Achieved **90%+ test coverage** by implementing comprehensive unit and integration tests with **pytest**, and created **MKDocs** documentation to ensure system reliability and long-term maintainability.
- Built a custom wrapper for **HTTP** and **gRPC** clients that monitors **500+** daily outgoing requests with **Prometheus** and built **Grafana** observability dashboards to reduce mean time to resolution **(MTTR)** by **60%**.

Software Developer Co-op - Incubation June 2025 – Aug 2025

Nokia *Ottawa, ON*

- Designed a **Python-based simulation tool** to model and validate a proof of concept for proactively managing **network traffic flow** in **5G networks**, simulating **7000+** concurrent packet distributions to proactively manage bottleneck flows.
- Implemented **parameterized unit tests** with **pytest**, expanding test coverage to **95%** and enhancing code robustness, scalability, and long-term maintainability.
- Collaborated in a cross-functional **SCRUM** team, actively participating in daily standups and sprint planning to ensure alignment between development goals and project deliverables.

Projects

CanadaChain - 3rd Place at Red Team Hackathon | *TypeScript, Docker, Next.js, TailwindCSS, Convex, React*

- Developed a full-stack cryptographic supply chain provenance system designed to verify product attestations in **TypeScript**, **Next.js**, **Convex**, and **React**.
- Implemented tamper-evident provenance using **SHA-256** content hashes and **Ed25519** supplier signatures on attestations, enabling independent **E2E** verification of Canadian content and Made in Canada / Product of Canada status.

Alsk | *JavaScript, React, LangChain, TailwindCSS, Pinecone, Express.js, MongoDB, Tavily, Node.js*

- Developed a Healthcare chat bot within a **MERN Stack** web application allowing users to chat about healthcare-related topics.
- Built a **Retrieval-Augmented Generation (RAG)** system using **LangChain**, **Tavily**, and **Pinecone** to enhance the chatbot's ability to provide accurate and context-aware responses.
- Designed a simple, user-friendly UI using **React** and **TailwindCSS**, and built **RESTful APIs** with **Express.js** to enable efficient client-server communication.